Arunabha Majumder

Mechanical Design Engineer

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EDUCATION

PhD, Mechatronics and Robotics
Aalborg University | 2023 - present
Thesis expected March

MTech, Mechatronics
AcSIR - CSIR-CMERI, India
2020 - 2022

BTech, Mechanical Engineering
Siddaganga Institute of Technology, India
2015 - 2019

Senior Diploma in Fine Arts (Painting)
Bangiya Sangeet Parishad
2008, First Division with Distinction

STUDY ABROAD

National University of Singapore
Sep - Nov 2025 (during my PhD)

SKILLS

CAD: SolidWorks (CSWA) 3D models, assemblies and technical drawings, tolerances, stiffness analysis in SolidWorks, Fusion 360 (basic, last used 2019)

Design: mechanisms, structural stiffness, torque and force calculations, worm gears, springs, GA optimisation

Build and test: machining (Bachelor's), 3D printing, assembly, test setups, actuators (motors, pneumatic muscles), encoders, force/torque sensors, Arduino, MATLAB/Simulink, LabVIEW

Art and design: hand sketching, painting, Adobe Illustrator, Photoshop, Procreate

ACHIEVEMENTS

Best Paper Award, IFToMM ISRM 2026
Best Major Project Award, 2019
TATA Technologies Prayatna grant
GATE qualified

LANGUAGES

English (C1), Bengali (native), Hindi

REFERENCES

Prof. Shaoping Bai
Aalborg University, sbh@mp.aau.dk

Assoc. Prof. Lars Rosgaard Jensen
Head of Section and Head of Labs, AAU

Assoc. Prof. Anderson de Souza Castelo Oliveira
Materials and Production, AAU

PROFILE

I am a mechanical design engineer who takes structural parts and assemblies from the first sketch to tested hardware. I model in SolidWorks, 3D print prototypes, have the final parts manufactured and verify them on physical test setups. For 6 years I have designed robots and exoskeletons, integrating motors and sensors and working closely with control and electronics engineers.

EXPERIENCE

Research Assistant

Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton: a variable stiffness mechanism (spring pretensioned by a worm gear) working in parallel with a 6 Nm motor; together they give the torque needed to lift the arm. It uses over 25% less motor energy (Best Paper Award, IFToMM ISRM 2026).
- Optimised the mechanism with a genetic algorithm (diameter, spring stiffness, pretension force), made the CAD and drawings in SolidWorks, 3D printed the first parts and had the final parts manufactured.
- Integrated the motor, encoder and force/torque sensors and measured its stiffness on a physical setup.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance. Force estimation used the variable stiffness actuators and the robot's Jacobian.
- Arm impedance study with the 5-bar robot (6 participants): Qualisys cameras recorded arm position data, and I implemented an Arduino-based trigger to sync the robot's motion with the recording.
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Soft Exosuit

CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit. Designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle (0.3 - 0.78% error).

Bachelor's Projects

Siddaganga Institute of Technology, India | *2017 - 2019*

- Built an arm exoskeleton and exo glove for a paralysed arm: CAD, machining, assembly and testing.
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

PUBLICATIONS

- Hybrid actuated shoulder exoskeleton, IFToMM ISRM 2026 (Springer), Best Paper Award.
- Neural network-based control of a pneumatic artificial muscle, IEEE CONECCT 2022.

PORTFOLIO

Videos and results from my work: arunabha22.github.io/arunabha-majumder